

Stage 2: Data Preparation and GARCH Volatility

Does GARCH Information Improve Forecasts of Implied Volatility?

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1 Data and sample

The study uses daily observations on the CBOE Volatility Index (VIX) and the S&P 500 Composite Index over **3 January 2012 to 21 August 2026**, giving **3,680 matched trading days**.

No single free provider covers both series: CBOE publishes the complete VIX history at no charge but does not publish S&P 500 index history. Fernandes et al. (2014) state no data source at all, and Wen et al. (2024) use OptionMetrics IvyDB and Tonghuashun—both paid subscriptions, and both options databases rather than index sources. Liu et al. (2015) face the same constraint, take both series from Yahoo Finance, and report that comparing Yahoo’s VIX against CBOE’s shows “only very minor differences for a few days.” Accordingly the VIX is taken from CBOE’s official file and the S&P 500 from Yahoo Finance (`^GSPC`); the Yahoo VIX was also downloaded, solely to replicate that validation on the present sample.

Across the 3,681 common dates the two VIX providers agree exactly, to two decimal places, on **99.76%** of observations; nine days differ by more than 0.01 and the correlation is 0.99997, reproducing the conclusion of Liu et al. (2015) on more recent data. The one material exception is 6 February 2026, where Yahoo reports 20.37 against CBOE’s 17.76. The S&P 500 rose 1.95% that day and CBOE’s path (21.77 $\rightarrow$ 17.76 $\rightarrow$ 17.36) is consistent with the rally while Yahoo’s is not, so the CBOE series is used throughout.

The calendars were compared before merging rather than letting an inner join discard mismatches silently. CBOE reports a VIX close on 32 dates on which the NYSE cash market was closed—US market holidays, concentrated from 2022 onward. These are genuine index values, since VIX is calculated from SPX option quotes disseminated in sessions that need not coincide with equity trading hours, but no matched return can be formed on those days and they are excluded. Neither series contains internal missing values or duplicate dates, and the largest gap between consecutive observations is the five-day NYSE closure for Hurricane Sandy ending 31 October 2012, left unimputed. Returns are computed as $r_t = 100 [\ln(P_t) - \ln(P_{t-1})]$ on the *full* price history before the sample is truncated, so the first in-sample return is measured against the 30 December 2011 close rather than being undefined.

2 Descriptive statistics

The mean of VIX exceeds its median, skewness is +2.86, and the maximum of 82.69 (16 March 2020) stands far above the minimum of 9.14: the index sits at a modest level most of the time and spikes sharply on occasion. Returns are the mirror image, with skewness of -0.64 , since the large moves that drive volatility upward are predominantly falls. Kurtosis near 19 for both, against a Gaussian value of 3, indicates pronounced fat tails, and Jarque–Bera rejects normality overwhelmingly.

3 Time-series properties

Persistence of VIX. The sample autocorrelation function (Figure 4) declines slowly:

Table 1: Descriptive statistics, 3 January 2012 – 21 August 2026 ($N = 3,680$). Kurtosis is unstandardised (normal = 3).

	VIX	S&P 500 return (%)
Mean	17.746	0.049
Median	16.180	0.068
Standard deviation	6.471	1.051
Minimum	9.140	−12.765
Maximum	82.690	9.089
Skewness	+2.863	−0.638
Kurtosis	18.939	19.314
Jarque–Bera (p)	$< 10^{-300}$	$< 10^{-300}$

Lag	1	5	10	22	66	125	250
ACF	0.961	0.854	0.751	0.563	0.337	0.262	0.046

The autocorrelation remains outside the ± 0.032 white-noise band as far as lag 250—a full trading year—and decays hyperbolically rather than geometrically. This is the long-memory behaviour documented by Fernandes et al. (2014), whose explanation applies here: VIX is a 30-calendar-day forward-looking measure sampled daily, so consecutive observations refer to heavily overlapping windows, which mechanically induces persistence. It is also why their plain HAR specification is so hard to beat at short horizons, and why Wen et al. (2024) find R^2 still improving at a 20-day lookback.

Stationarity. The Augmented Dickey–Fuller test rejects the unit-root null decisively for VIX (ADF = -5.560 , $p = 1.5 \times 10^{-6}$) and for returns (ADF = -13.420 , $p = 4.2 \times 10^{-25}$). VIX is therefore stationary but strongly persistent: shocks do die out, but slowly.

Volatility clustering. The contrast between the autocorrelation of returns and of squared returns is decisive (Figure 5):

Lag	1	2	5	10	22
ACF of r_t	−0.116	+0.072	+0.034	−0.034	−0.078
ACF of r_t^2	+0.449	+0.469	+0.290	+0.218	+0.065

Raw returns are close to unforecastable, with small autocorrelations that alternate in sign; squared returns are not, remaining positive throughout. The direction of tomorrow’s move is essentially unpredictable, its magnitude highly predictable from recent magnitudes. Ljung–Box gives $Q(10) = 3,955$ on squared returns against 236 on raw returns, and Engle’s ARCH-LM test returns LM = 1,109 at five lags with $p \approx 10^{-237}$. These results establish the precondition for a conditional-variance model.

4 GARCH(1,1) estimation

The model estimated on S&P 500 log returns is

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

All three variance parameters are significant at any conventional level, with $\alpha, \beta > 0$ and $\alpha + \beta < 1$, so the non-negativity and stationarity conditions hold. $\omega = 0.0431$ is the constant floor of the variance equation; combined with the persistence it implies a long-run daily variance of $\omega/(1 - \alpha - \beta) = 1.010$ and an annualised long-run volatility of 15.95, close to the sample standard deviation of returns as it should be. $\alpha = 0.167$ is the reaction coefficient—an unexpected move raises today’s variance by

Table 2: GARCH(1,1) with Gaussian innovations, estimated by maximum likelihood with robust standard errors.

Parameter	Estimate	Std. error	<i>t</i> -statistic
ω	0.0431	0.0093	4.63
α	0.1666	0.0230	7.24
β	0.7907	0.0256	30.92
$\alpha + \beta$	0.9573		
Half-life of a variance shock	15.9 trading days		
Long-run annualised volatility	15.95		

16.7% of the squared surprise—while $\beta = 0.791$ is the carry-over from yesterday’s variance, the most precisely estimated parameter in the model ($t = 30.9$) and the one that dominates the dynamics.

$\alpha + \beta = 0.9573$ is the quantity of interest. It lies below unity, so variance is mean-reverting, but is high enough that shocks decay slowly: the half-life is $\ln(0.5)/\ln(0.9573) \approx 15.9$ trading days, roughly three trading weeks for half of a shock to dissipate. This is the standard result for daily equity index returns, where $\alpha + \beta$ typically lies between 0.95 and 0.99, and it mirrors the persistence already documented for VIX itself. Given return kurtosis of 19.3 the model was re-estimated with Student- t innovations: α is essentially unchanged at 0.167, persistence rises to 0.984 with $\nu = 5.48$, and BIC prefers that specification decisively (9,019 against 9,247). The persistence conclusion therefore does not depend on the distributional assumption; the Gaussian model is retained because the brief specifies a basic GARCH(1,1).

5 VIX and GARCH volatility compared

The conditional volatility series is returned in daily percentage units and is annualised as $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$, placing it on the same scale as VIX. Figure 6 plots the two together.

VIX exceeds annualised GARCH volatility on **86.7%** of days, by an average of **3.19 volatility points—about 22% of the GARCH mean** (14.56 against a VIX mean of 17.75). A discrepancy this one-sided is not estimation error but the **variance risk premium**: VIX is a risk-neutral expectation extracted from option prices and embeds compensation demanded for bearing volatility risk, whereas GARCH, estimated from realised returns alone, delivers only the physical forecast. Liu et al. (2015) report their GARCH-computed VIX running 20–30% below the market VIX before September 2003 and 10–13% below afterwards, interpreting precisely that shortfall as a variance risk premium; the 22% found here on 2012–2026 data falls in the same range. The exception is instructive: GARCH exceeds VIX on 13.3% of days, and the largest overshoot—44 points on 17 March 2020—follows directly from the -12.77% return of 16 March, since σ_t^2 depends on ϵ_{t-1}^2 and a single extreme return is mechanically propagated into the next day’s variance. Annualised GARCH volatility reached 120 while VIX, already pricing an eventual return to calm, stood at 75.9.

In levels the two series move together closely: Pearson 0.833, Spearman 0.730, and 0.793 in logarithms, stable across sub-periods and strongest during 2020–2022 (0.889). In daily *changes*, however, the correlation collapses to -0.007 . The lead-lag profile resolves the apparent conflict:

k	−2	−1	0	+1	+2
$\text{corr}(\Delta\sigma_{t+k}^{\text{GARCH}}, \Delta\text{VIX}_t)$	+0.002	+0.026	−0.007	+0.410	+0.035

The correlation is near zero contemporaneously and +0.41 at a one-day lag. The reason is structural: σ_t is a function of information dated $t-1$ and cannot respond to today’s news until tomorrow, whereas VIX is priced off current option quotes and responds immediately. The two measures agree about the *level* of volatility but learn about *changes* one day apart. This bears directly on the project: a predictor that merely tracked VIX would be redundant, whereas GARCH volatility is strongly

related in levels, reacts on a different schedule, and differs by a systematic risk-premium wedge that VIX contains by construction and GARCH cannot. Whether that survives as *incremental* predictive power once VIX’s own history is conditioned upon is the question for the forecasting stage, and the ACF result above shows the own-history baseline to be a demanding one.

6 Conclusions

Is VIX highly persistent? Yes. First-order autocorrelation is 0.961, still 0.563 after one month and 0.337 after a quarter, with the ACF outside the white-noise band as far as lag 250. Decay is hyperbolic rather than geometric. VIX is nonetheless stationary ($\text{ADF} = -5.560$, $p = 1.5 \times 10^{-6}$).

Is there volatility clustering in S&P 500 returns? Yes, unambiguously. Raw returns are near-unforecastable, but squared returns are autocorrelated at 0.449 at lag 1 and remain positive beyond lag 20. Ljung–Box gives $Q(10) = 3,955$ on squared returns against 236 on raw returns, and ARCH-LM returns $\text{LM} = 1,109$ with $p \approx 10^{-237}$.

Is GARCH volatility highly persistent? Yes. $\alpha + \beta = 0.9573$ implies a half-life of 15.9 trading days. The process is stationary, with a long-run annualised volatility of 15.95. Under Student- t errors persistence rises to 0.984.

How closely do VIX and GARCH volatility move together? Closely in levels (Pearson 0.833), hardly at all in daily changes (-0.007). The lead–lag correlation of $+0.410$ at one day shows that GARCH responds to the same information one day later, because σ_t conditions on data through $t - 1$ while VIX is priced off current quotes.

What is the main difference between VIX and GARCH volatility? They are different objects rather than competing estimates of the same one. VIX is forward-looking, extracted from option prices, and expresses the market’s expectation of volatility over the coming 30 calendar days, whereas GARCH volatility is backward-looking and computed mechanically from the history of realised returns. VIX is risk-neutral and therefore includes a variance risk premium, evident in the 22% average wedge; GARCH is a physical-measure forecast and includes none. VIX also reprices within the trading day while GARCH cannot move until the day after a shock—the source of both the one-day lag in the change correlation and the March 2020 episode, in which GARCH extrapolated the crash to an annualised 120 while VIX peaked at 82.7.

References

- Fernandes, M., Medeiros, M. C., & Scharth, M. (2014). Modeling and predicting the CBOE market volatility index. *Journal of Banking & Finance*, 40, 1–10.
- Liu, Q., Guo, S., & Qiao, G. (2015). VIX forecasting and variance risk premium: A new GARCH approach. *North American Journal of Economics and Finance*, 34, 314–322.
- Wen, C., Zhai, J., Wang, Y., & Cao, Y. (2024). Implied volatility is (almost) past-dependent: Linear vs non-linear models. *International Review of Financial Analysis*, 95, 103406.

Appendix: Figures

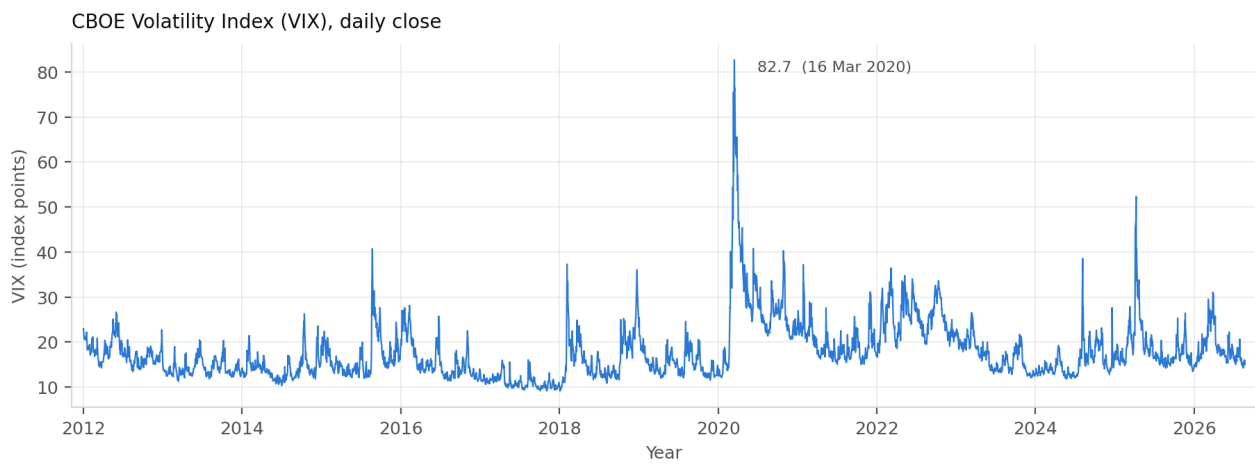


Figure 1: VIX, daily close, 2012–2026. Source: CBOE.

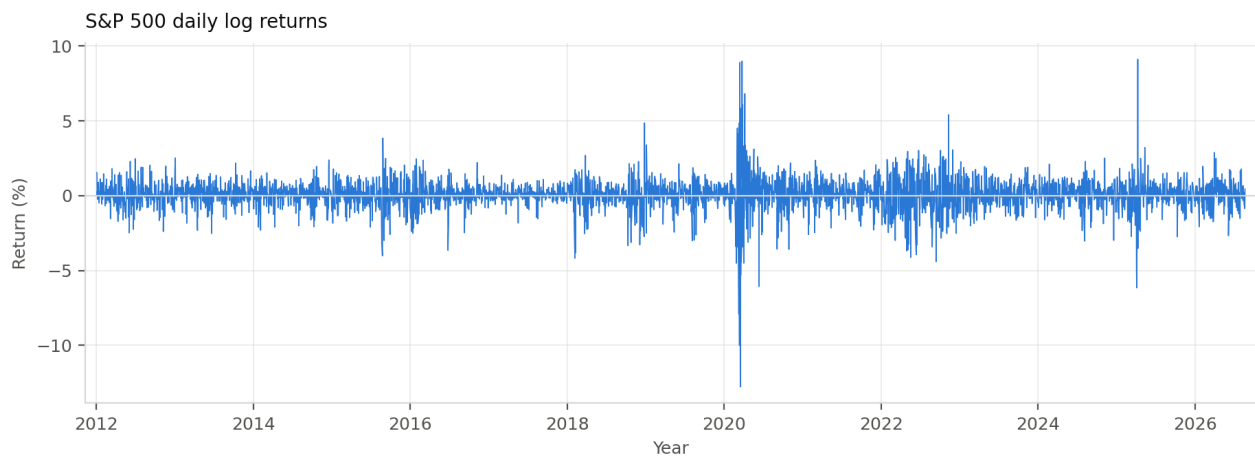


Figure 2: S&P 500 daily log returns. Amplitude varies systematically over time.

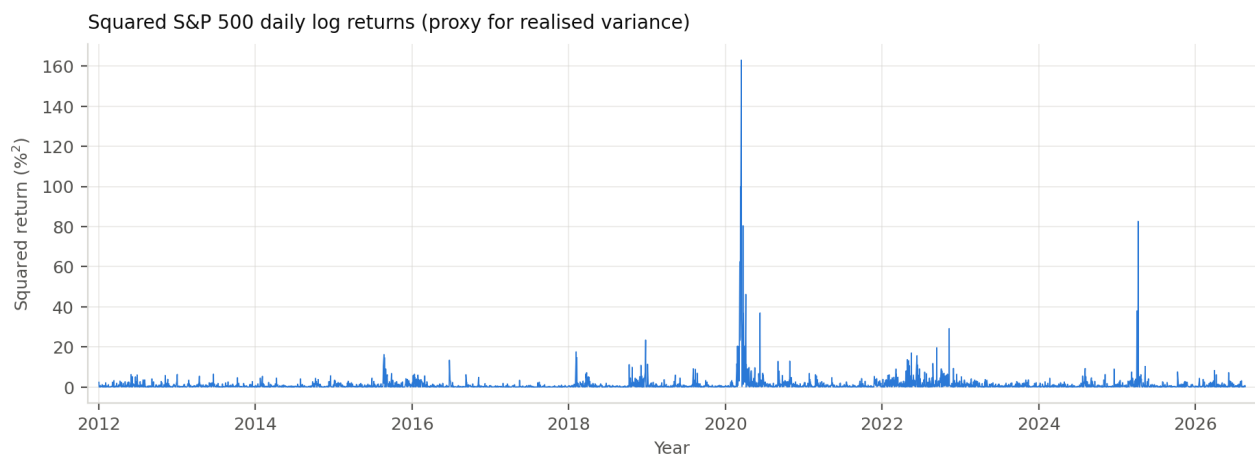


Figure 3: Squared S&P 500 daily log returns. Removing the sign makes the clustering of large moves unmistakable.

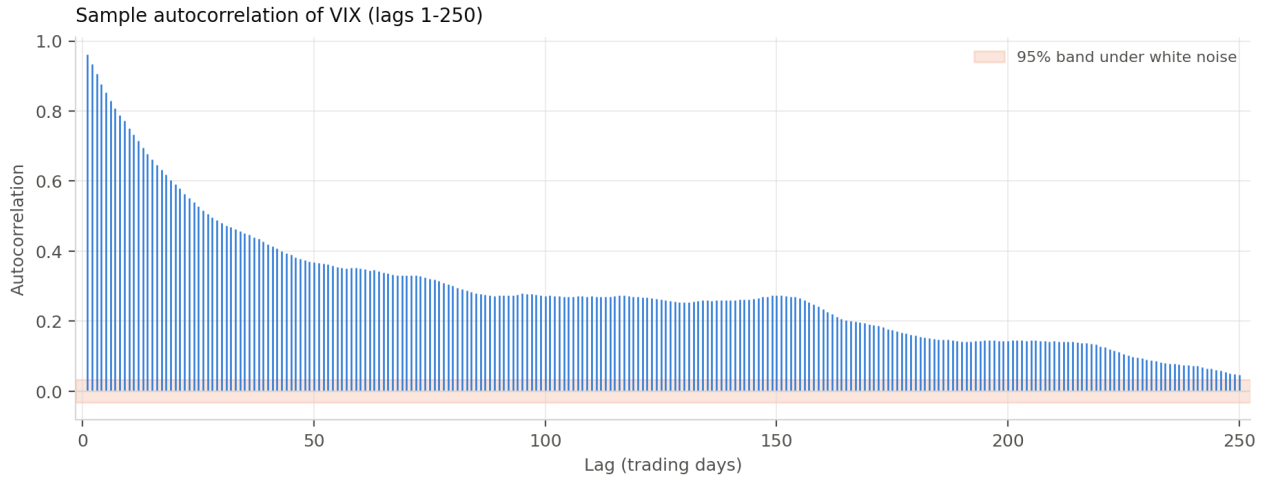


Figure 4: Sample autocorrelation of VIX to lag 250. The ACF remains outside the 95% white-noise band across the entire range.

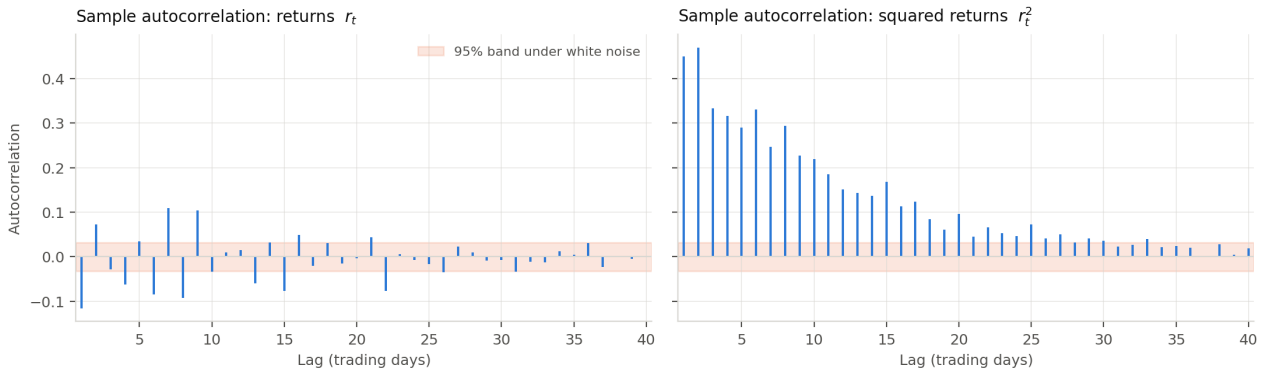


Figure 5: Sample autocorrelation of returns (left) and squared returns (right). Direction is unforecastable; magnitude is not.

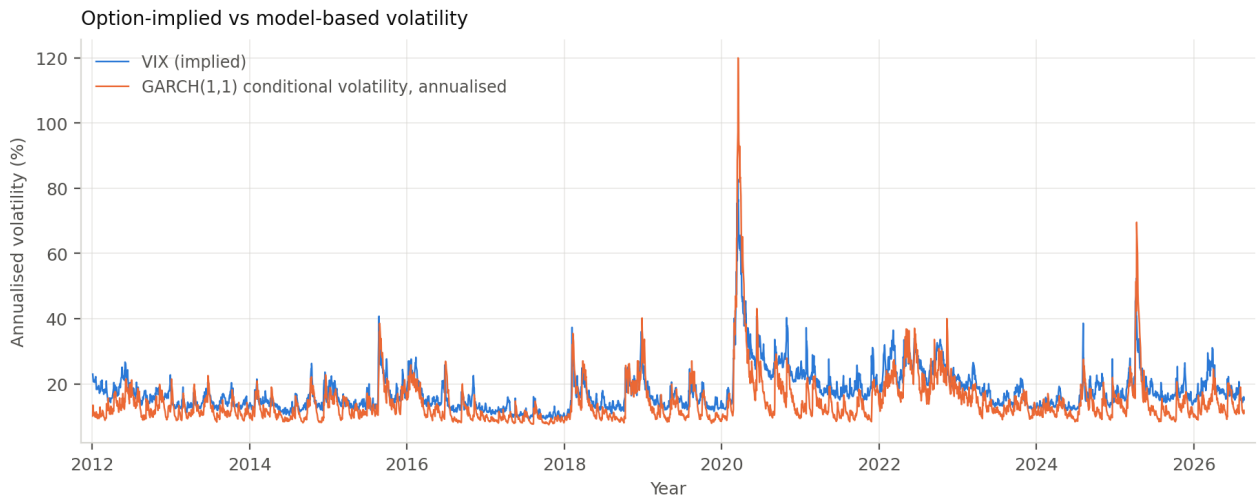


Figure 6: VIX against annualised GARCH(1,1) conditional volatility. Both series are in annualised volatility points and share a single axis.

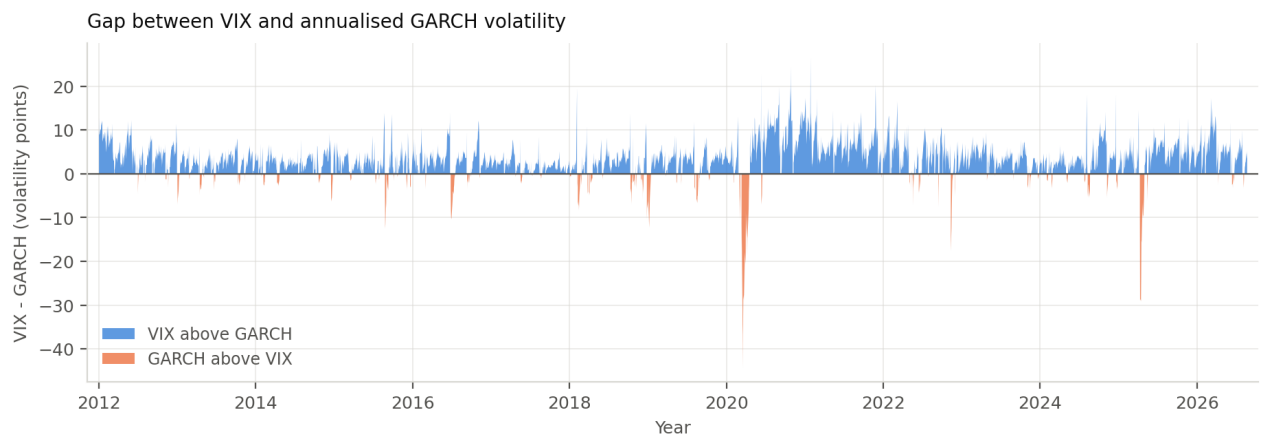


Figure 7: VIX minus annualised GARCH volatility. The predominance of positive values is the variance risk premium; the negative spikes follow extreme single-day returns.